

NATIONAL TELE-OPHTHALMOLOGY AI SCREENING PROGRAM

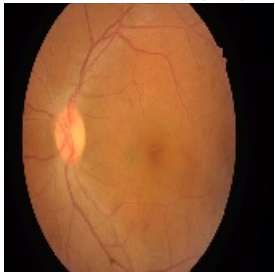
Explainable AI Clinical Decision Support System | Rural PHC Triage (SIH PS 26038)

Patient ID: IND-PHC-2026-0814 **Age/Sex:** 58 / Male **Screening Date:** 17-Sep-2026 23:55
PHC Health Centre: PHC Mulbagal, Kolar District #04 **Frontline Operator:** ASHA Health Worker #108
Eye Evaluated: Right Eye (OD) (Nasal: Left) **QA Usability Score:** 97/100 [PASS]

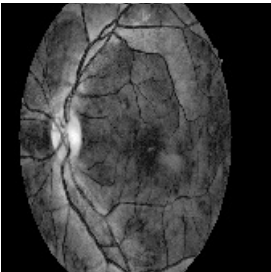
TRIAGE ACTION: NON-REFERABLE: ROUTINE ANNUAL SCREENING

No sight-threatening diabetic retinopathy detected. Schedule follow-up in 12 months at PHC.

1. Raw Fundus (OD)



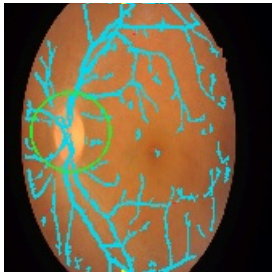
2. Enhanced Green



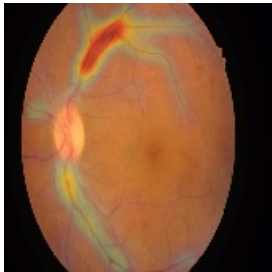
3. Retinal Vessels



4. Lesion Detections



5. Grad-CAM XAI

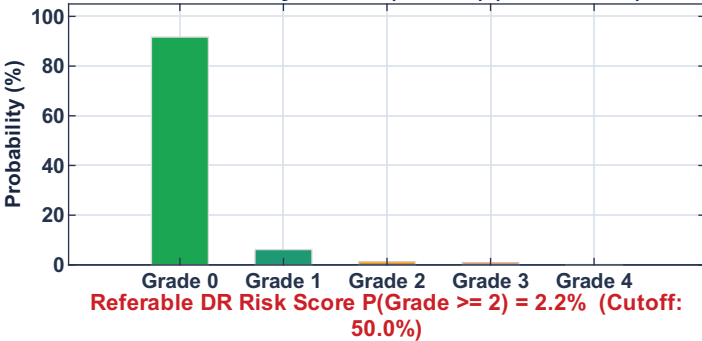


Legend: [Yellow]: Hard Exudates | [Ruby Red]: Microaneurysms & Hemorrhages | [Heatmap]: Deep Feature Attention (Grad-CAM)

QUANTITATIVE BIOMARKERS & ETDRS 4-2-1 CLINICAL AUDIT

ETDRS Staging: MODERATE NPDR (Lesions present in 2-3 quadrants)
ETDRS 4-2-1 Rule: 4Q Hemo: [0/4] | VB Quads: [0] | IRMA: [0]
Distance to Fovea (CSME): 3988 μm (2.66 DD) — Low
Hard Exudates (Bright): 1 clusters (21 px area)
Microaneurysms / Blot Hemo: 4 MAs | 0 Hemorrhages
Vascular Density: 11.38% FOV **Optic Disc:** [128, 260] (Left)
Systemic EHR Profile: T2D: 11 yrs | Last HbA1c: 8.9% | BP: 138 / 88 mmHg (Moderately Controlled)
Last Encounter Record: 18-Jan-2026 (8 months ago) (Grade: Grade 1 (Mild NPDR - Isolated microaneurysms))

AI DR Severity: No DR (Normal) (Conf: 91.7%)



RECOMMENDED REFERRAL CARE: PHC Mulbagal, Kolar District #04 - Vision Sub-Centre [Level 1: Primary Health Centre (PHC)]

Est. Distance: ~2.5 km | **Timeline:** ROUTINE ANNUAL RESCREEN (12 Months) | **Helpline:** 104 (National Health Helpline)

OPHTHALMOLOGIST TELEMEDICINE REVIEW & VALIDATION (Target Review Time: under 30 Seconds)

Clinical Impression: [] Concordant with AI diagnosis [] Upgrade Severity [] Downgrade / Artefact
Prescribed Action: [] Tele-Consultation Booked [] Anti-VEGF / Laser Referral [] PHC Annual Rescreen
Reviewing Ophthalmologist Signature: Date/Time: